

PRS Cross-Ancestry Portability & Recalibration Benchmark

Case Study Summary

THE PROBLEM

Polygenic risk scores (PRS) trained on predominantly European-ancestry GWAS are well documented to underperform when applied to other ancestries. This project measures that gap with a controlled, synthetic ground truth on real 1000 Genomes multi-ancestry genotypes (GCTA-simulated phenotype, known causal variants and heritability) — and checks the same pattern against a real, published PGS Catalog score.

HEADLINE FINDING

A simple clumping+thresholding PRS (PRSice-2), tuned to its best EUR threshold, beats both Bayesian genome-wide methods (PRS-CSx, LDpred2-auto) in the EUR held-out set — but loses to PRS-CSx in every single one of 8 non-EUR (ancestry × scenario) comparisons, and to LDpred2-auto in 6 of 8. No method wins universally (see Scope & Limitations):

Scenario	Ancestry	PRSice-2 (best)	PRS-CSx	LDpred2-auto
1 (equal)	AFR	0.0077	0.0275	0.0117
1	AMR	0.0098	0.0594	0.0136
1	EAS	0.0008	0.0165	0.0015
1	SAS	0.0080	0.0279	0.0118
2 (varying)	AFR	0.0081	0.0230	0.0130
2	AMR	0.0009	0.0480	0.0065
2	EAS	0.0090	0.0238	0.0251
2	SAS	0.0161	0.0411	0.0112

R² against the true simulated phenotype. PRS-CSx pattern holds in all 8/8 non-EUR comparisons; LDpred2-auto in 6/8.

RECALIBRATION: WHAT IT FIXES AND WHAT IT DOESN'T

Empirical per-ancestry recentring/rescaling recovers 100% of calibration (mean/scale matched to the EUR reference in every case) and 0% of discriminative accuracy (R² identical before/after to 4 decimal places, in all 95 result rows) — by mathematical necessity, since linear recentring cannot change a Pearson correlation. This makes a distinction the field warns is often conflated concrete rather than asserted.

REAL-SCORE CROSS-CHECK (DESCRIPTIVE ONLY)

A real published height PGS (PGS000297) applied to the same genotypes: 0 of 5 super-populations' score ranking matches published national height statistics — a real, small-scale illustration of the same portability problem, not a validated accuracy claim (1000 Genomes has no phenotype data to validate against).

SCOPE & LIMITATIONS

chr21+chr22 smoke-test scope (not genome-wide); LDpred2-auto's LD reference is chr22-only, narrower than PRS-CSx's official multi-population panel, a plausible driver of its weaker cross-ancestry showing above; single simulation replicate; small per-ancestry sample sizes. Full detail: METHODS.md and reports/final_writeup/TECHNICAL_WRITEUP.md.